# An Analysis of Spatial Clustering in MLB Batted Ball Distributions

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SOSC 13320
Spatial Analysis III
Spring 2026

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# Introduction

Understanding how Major League Baseball (MLB) players hits are distributed across the diamond is central to evaluating defensive strategies. While traditional statistics summarize performance outcomes, they often overlook the spatial dimension of hitting.

This project investigates whether MLB hitters exhibit spatial clustering in their batted ball locations, and how these patterns of concentration vary among a selection of high-volume hitters. We are using data from the 2023 MLB regular season. High-volume hitters will be defined as any player with a minimum of 500 plate appearances and a minimum of 100 batted balls in play.

To implement this analysis, we will employ spatial clustering methods such as density-based clustering or kernel density estimation to identify regions of high batted ball concentration. Then we will assess statistical significance through a series of permutations, determining whether clustering patterns deviate from randomness.

# Literature Background

The increasing availability of high-resolution tracking data has transformed the metrics and analysis of performance in Major League Baseball. Statcast, as described by the MLB is a “state-of-the-art tracking technology” that enables researchers to investigate aspects of gameplay beyond the traditional box scores “in ways that were never possible in the past” (MLB 2026). Some of these granular aspects include swing mechanics, and ball flight. Statcast was originally implemented in 2015 consisting of a combination of cameras and radar systems, however in 2020, the MLB added the Hawk-Eye, a high-speed camera allowing for a replay system and various new features.

One substantial body of work examines batting performance and offensive prediction, leveraging Statcast data including “launch angle, launch velocity, and hit distance” (Bailey 2017). Using logistic regression to estimate the hit probabilities, these predictions are then aggregated to forecast future batting averages. Importantly, Bailey’s analysis utilized Statcast data in combination with Player Empirical Comparison and Optimization Test Algorithm (PECOTA), a prediction system that considers previous seasons to predict batting averages. A linear regression was conducted on the relationship between the previous PECOTA and new Statcast data to create an even stronger metric for predicting batting averages.

Gergis (2024) provides a recent and comprehensive analysis of on-base percentage, with a machine learning approach and 2015-2023 Statcast data. The analysis leverages over one million pitch observations, and compares traditional statistical methods such as logistic regression, to the more advanced techniques of XGBoost, a machine learning library. They found a substantial performance advantage for XGBoost, notably a reduced feature set containing fewer than one-third of the original variables, can achieve a nearly identical predictive performance.

Machine learning applications have been further extended to optimization in outfield defensive positioning. A study analyzing over fifty-one thousand batted balls from the 2023 MLB season evaluates how outfield positioning influences defensive outcomes. Results show that for ever one foot increase in straight line distance between a batted ball and the fielder, the odds of a hit increase by about five percent. Hwang (2025) also found that higher exit velocity and steeper launch angles are associated with a lower probability of a base hit. However, their findings also suggest that MLB outfielder positions are already near-optimal in most cases. The minimal difference in machine-learning metrics and actual defensive positions show a convergence towards a positional equilibrium. Teams independently arrive at similar spatial strategies.

A recent study using large language models to generate baseball spray charts. Using collegiate play-by-play text, they constructed approximately ninety-five percent accurate spray charts when compared to video tracking. The study finds that aggregated spatial patterns are highly stable despite noisy inputs (Michielssen 2024). These results introduce the idea that hitter outcomes are not randomly distributed across the field, but rather structured in spatial density patterns. While spray charts are effective in summarizing batted ball tendencies, they aggregate continuous spatial variation into discrete regions of the field, limiting their ability to provide meaningful results when it comes to if their exists actual statistically significant clustering. Therefore, there is motivation to apply clustering methods of spatial analysis to batted ball locations in order move beyond simple descriptive visualization.

A complementary line of research suggests a spatial structure in baseball data, when observations are grouped rather than analyzed in isolation. Marcou (2020) focuses on pitchers and the quality of contact they allow. He finds that while individual metrics such as exit velocity show substantial variability, clustering reveals that pitches tend to fall into stable groups. For instance, sinker-heavy pitchers exhibit ground ball rates near fifty-nine percent, whereas other pitcher profiles show roughly forty percent ground ball rates. Building on this idea, our project shifts the analysis from pitches to batters themselves. By applying spatial clustering methods, we aim to determine whether hitters exhibit similar latent profiles in terms of where they put the ball in play.

Oda and Hirotsu (2026) extend on Marcou’s research as they develop a framework for classifying fielders in the Nippon Professional Baseball league using Gaussian model clustering. Using a dataset of one-hundred and fifteen hitting-related variables over three-hundred and twenty-seven players. They reduce multidimensional via principal component analysis, filtering for components that explain about seventy percent of the variance in performance categories related to plate discipline and win probability. This study finds that players grouped within the same cluster exhibit similar latent performance profiles. The authors multidimensional approach captures nuances that are overlooked by individual metrics. Our project moves the clustering work done by Oda from feature space to the physical space. The core argument that one-hundred and fifteen hitting indices can be condensed into distinct latent dimensions supports our aggregation. The spatial clustering we aim to perform reveals latent geographic structure in batted-ball locations.

Cox (2010) provides a direct methodological precedent for spatial analysis of batted-ball locations in the MLB. Cox performs a spatial clustering of batted ball locations, through a hot spot analysis and Moran's I for significance. Cox examines manually collected data from the 2009 season at the Texas Rangers ballpark, pre-statcast. She goes on to perform density surface estimation and test for autocorrelation. This analysis lays the groundwork for our project. Her analysis was confined to a single ballpark and balls that left the field of play, however she notes that the framework could be extended to any batted-balls. Our project applies spatial clustering methods to all batted-balls in play across a stratified sample. This study lays the groundwork for methodology that we aim to extend to a different question using more updated statcast data.

Another study that builds on the spatial structure in batting outcomes examines batting ability as a spatial Gaussian field with isotropic covariance. Cross and Sylvan (2015) empirically demonstrate that batting performance exhibits spatial dependence. They demonstrate through Monte Carlo simulations that traditional grid-based heat maps produce inaccurate estimate of batter ability. They found that a spatial analysis substantially outperforms discrete summaries. This study demonstrates the methodological value of treating space continuously, a vital component to our analysis. Extending this line of work, Comiskey (2017) develops variable-resolution heat maps that adapt to changing density across the strike zone. This analysis addressed the computational bottleneck that Bayesian spatial models face with batted-ball data. Combined, these works astablish batting outcomes requiring methods beyond discrete summarization. This study directly reinforces our analysis choice aims to utilize Kernal Density Estimations over fixed grids. Our point-pattern and clustering approach serve as a complementary methodology.

Kaji and Sylvan (2023) apply point pattern analysis to MLB batted-ball locations to model the inhomogeneous Poisson process underlying batted-ball distributions. They use statcast data and compare hitters by handedness. This study explores kernel density smoothing with optimal bandwidth selection, producing heat maps that visually distinguish the spatial patterns of right-handed and left-handed hitters. They only test three players, whereas our study expands this to fifteen players. Our study further builds directly on this framework but extends it to a different stratification of hitters, density based clustering, and significance testing to assess clustering patterns and if or how they deviate from spatial randomness.

Hochstedler (2016) explores the application of spatiotemporal machine learning techniques in professional sports. Methods such as Voronoi tessellations and neural networks demonstrate how spatial data can inform decision-making in sports. His work highlights the importance of spatial structure in player and ball movement, however largely assumes the existence of meaningful spatial patterns. Building on this, our project seeks to formally test whether these spatial structures exist in batted-ball distributions, and further test the significance of these spatial structures.

A complementary line of research emphasizes the importance of modeling batted-ball outcomes independently of external factors such as defense or varying ballparks. Healey (2017) develops a Bayesian model that uses kernel density estimation to construct a continuous mapping from batted-ball parameters including exit velocity and launch angle. Healey's use of nonparametric density estimation reinforces the need to treat batted-alls as a continuous spatial phenomenon. Our analysis recognizes that meaningful interpretation of our clusters requires accounting for underlying contact and other variables, outside of simply observed outcomes.

Young (2020) models the spatial distribution of batted-balls directly through spray-chart densities. They develop a Synthetic Estimate Average Matchup (SEAM) methodology which estimates the conditional distribution of balls-in-play for batter-pitcher matchups. This study incorporates player characteristics across similar players to model the structured probability distributions of different matchups. Young highlights the importance of density estimation techniques in capturing these patterns. Our analysis shifts this study from estimating conditional spray distributions to identifying spatial clustering. Extending on this, an earlier line of work focuses on the visualization of spatial baseball data, and how batted-ball data can advance scouting reports. Superak (2011) represents batted-ball data through density plots and contour maps to identify patterns in hitting behavior. One route in which we aim to take this project is to examine how we can apply our spatial analysis into real-world decision-making such as optimizing defensive positioning or tailoring strategic alignments to hitter-specific tendencies.

## Data and Methodology

To arrive at the 15 players of interest for this study, MLB players with at least 500 plate appearances (PAs) in the 2023 season were identified. 136 players met this criteria. Then batted ball dispersion data for each player was collected, and this data captures how frequently each player hits the ball to each field direction. The metrics of interest were pull percentage (pull%), opposite-field percentage (oppo%), and straight ahead percentage (cent%). Then each player was classified into one of four archetypes: balanced, center, opposite-field (oppo), or pull. If a player's pull%, oppo%, or cent% is larger than 40% they were classified into the associated archetype of the direction the tend to hit the most. If a player's pull%, oppo%, and cent% were all less than 40% this hitter would be classified as a balanced hitter.

75 of the players were classified as pull hitters, 51 as balanced hitters, 9 as center hitters, and 1 as an opposite field hitter. A stratified sampling approach was then used to ensure representation across all player archetypes. Based on proportional allocation from the full sample of 136 players, the initial values were approximately 8.27 pull hitters, 5.63 balanced hitters, 0.99 center hitters, and 0.11 opposite-field hitters given a sample size of 15 total players.

These values were then rounded to produce the final sample. Center and opposite-field hitters were both rounded to 1 ensuring that each archetype is represented in the sample. This rounding occurred because excluding the smaller groups would eliminate meaningful variation in hitting tendencies. Pull and balanced hitters were rounded down to 8 and 5 to maintain the total sample size of 15 while keeping the distribution as close as possible to the original proportions.

A final sample size of fifteen was chosen to balance representativeness of the overall 136 player population and to ensure the dataset remained manageable for spatial analysis.

The fifteen players chosen are as follows. Pull Hitters: Tyler Stephenson-R, Whit Merrifield-R, Jake Cronenworth-L, Seiya Suzuki-R, CJ Abrams-L, Josh Lowe-L, Josh Jung-R, Teoscar Hernández-R. Balanced hitters: Trent Grisham-L, Pete Alonso-R, Eugenio Suárez-R, TJ Friedl-L, J.T. Realmuto-R. Center Hitter: Yandy Díaz-R. Opposite-Field Hitter: Bo Bichette-R.

The side from which a player bats was not considered in constructing the sampling method. Nevertheless, the resulting sample which consisted of 5 left-handed batters and 10 right-handed batters approximately reflects the overall distribution of handedness in Major League Baseball of 30.3% (Grondin et al., 1999, as cited in Nachtigal & Barnes, 2019).